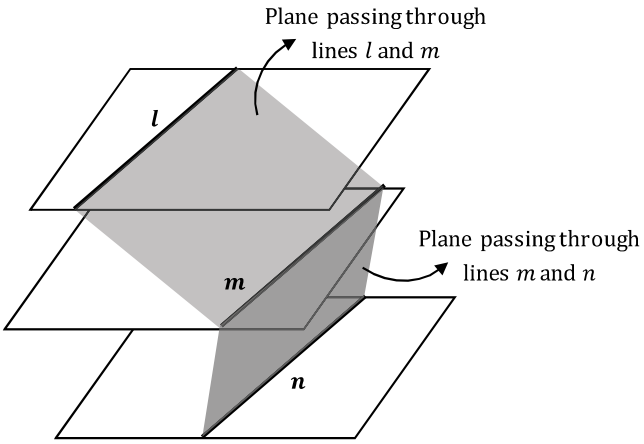
5.17 Geometry–Lines & Triangles

207. Working with the statements:

Statement I:

We can have a situation as shown below:



The lines l, m and n are parallel, however, the plane passing through l and m does not pass through n, implying that the lines are in different planes.

Thus, statement I is incorrect.

Statement II:

Since both l and m are separately parallel to n, the lines l and m must be parallel to each other as well.

Thus, statement II is correct.

Statement III:

From the above diagram, it is clear that lines l and m do not need to be the same line.

Thus, statement III is incorrect.

The correct answer is option B.

208. We know that:

$$AB = \frac{1}{2}CD$$

$$=> CD = 2AB \ldots (i)$$

 © 1999–2016 Manhattan Review

$BD = \frac{3}{2}AC \ldots$ (ii)

$BC = 24 \ldots$ (iii)

From (ii):

$\Rightarrow BC + CD = \frac{3}{2}(AB + BC)$

Using (i) and (iii):

$\Rightarrow 24 + 2AB = \frac{3}{2}(AB + 24)$

$\Rightarrow 48 + 4AB = 3AB + 72$

$\Rightarrow AB = 24$

$\Rightarrow CD = 2 * 24 = 48$

$\Rightarrow AD = AB + BC + CD$

$= 24 + 24 + 48$

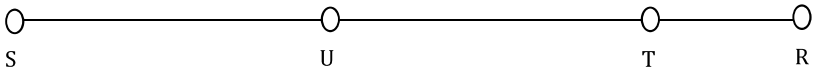
$= 96$

The correct answer is option D.

209. Since the length of RS > the length of RT, R must be closer to T than to S.

Also, since sum of the lengths of RS and RT is greater than the length of ST, R must line beyond the line segment ST.

The sequence of the points R, S, T and U is shown in the diagram below:



Since U is the midpoint of ST, we have:

$SU = UT = \frac{ST}{2}$

$= \frac{16}{2}$

$= 8$

RU = RT + TU

= 4 + 8

= 12

The correct answer is option C.

210. Here we know that both dressing rooms are equidistant from a cash register and distance between these 2 dressing tables is constant that is 16. So if we join all these 3 places with each other, it will give us an isosceles triangle.

So here property about lengths of sides of triangle is "Sum of any two sides is always greater than third side and positive difference between any two sides is always less than third side."

• Case I – Isosceles triangle is of sides: 6, 6, 16. Here $6 + 6 \not> 16$ so such a triangle does not exist. So this is NOT the possible case.

• Case II – Isosceles triangle is of sides: 12, 12, and 16. This triangle follows above mentioned property. So this is a possible case.

• Case III – Isosceles triangle is of sides: 24, 24, and 16. This triangle follows above mentioned property. So this is a possible case.

The Correct answer is option E.

211. Given that:

$x < y < z$

Since z is the longest side in the right triangle, it must be the hypotenuse.

Also, x and y are the perpendicular legs of this right angled triangle.

Here let us recall formula to find the area of Right Angled Triangle:

$= \frac{1}{2} *$ Product of two legs

Thus, area of the triangle:

$= \frac{1}{2} * x * y = 1$

$=> xy = 2$

If we assume that $x = y$, we have:

 © 1999–2016 Manhattan Review

$x = y = \sqrt{2}$

However, it is known that $y > x$, we can conclude that $x < \sqrt{2}$ and $y > \sqrt{2}$.

$=> y > \sqrt{2}$

The correct answer is option A.

212. Since z is the longest side in the right triangle, it must be the hypotenuse.

Thus, we have:

$x^2 + y^2 = z^2 \ldots$(i)

Also, x and y are the perpendicular legs of the triangle.

Thus, area of the triangle:

$= \frac{1}{2} * x * y = 1$

$=> xy = 2 \ldots$(ii)

From (i), we have:

$z^2 = x^2 + y^2$
$= (x + y)^2 - 2xy$
$= (x + y)^2 - 4$ (since $xy = 2$, from (ii))

Given $xy = 2$, the minimum value of $(x + y)$ occurs if $x = y$.

Let us verify:

Say, $x \neq y$:

$x = 2, y = 1 => x + y = 3$

However, if $x = y$:

$x = y = \sqrt{2} => x + y = 2\sqrt{2} \sim 2 * 1.4 = 2.8 < 3$

However, the maximum value of $(x + y)$ is undefined and tends to infinity.

Say, we take $y = 100, x = 0.02 => x + y = 100.2$

Thus, the value of $(x + y)$ can be increased to any arbitrarily high value.

Thus, we can see that given $xy = 2$, $(x + y)$ has only a minimum value of $2\sqrt{2}$.

© 1999–2016 Manhattan Review

Hence, we have:

$z^2 = (x + y)^2 - 4$

=> The minimum value of z^2 occurs when $(x + y)^2$ is minimum i.e. $(x + y)$ is minimum

=> Minimum value of $z^2 = \left(2\sqrt{2}\right)^2 - 4 = 4$

=> $z > 2$ (the equality does not hold since x and y cannot be equal since $x < y$)

The correct answer is option A.

213. Since DA = DC, triangle DAC is isosceles

=> $\angle DAC = \angle DCA = 30^o$

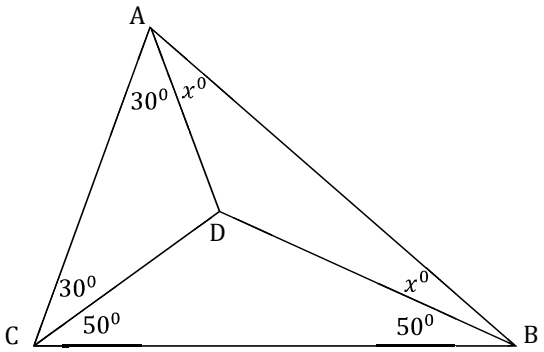
Since DB = DC, triangle DBC is isosceles

=> $\angle DBC = \angle DCB = 50^o$

Since DA = DB, triangle DAB is isosceles

=> $\angle DBA = \angle DAB = x^o$

 © 1999–2016 Manhattan Review



Since sum of the internal angles in a triangle is 180°, we have:

$(30 + x) + (x + 50) + (50 + 30) = 180$

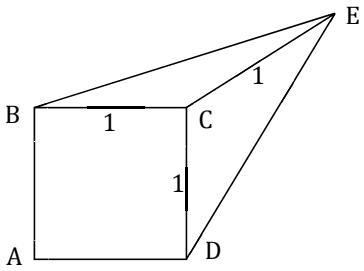
$=> 2x = 20$

$=> x = 10$

The correct answer is option A.

214.

© 1999–2016 Manhattan Review



We observe that triangles BCE and DCE are isosceles.

Also, triangle BCE ≅ triangle DCE

(Since CE is common, BC = CD and BE = DE)

Thus, $\angle BCE = \angle DCE$

$= \frac{360^o - 90^o}{2} = 135^o$

Considering triangle BCE alone, we have:

In triangle CEF:

$\angle ECF = 180^o - 135^o = 45^o$

$=> \angle CEF = 180^o - (90^o + 45^o) = 45^o$

Thus, triangle CEF is isosceles.

Let CF = EF = x

Thus, from Pythagoras' theorem in triangle CEF:

$x^2 + x^2 = 1^2$

$=>$ EF $= x = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

Thus, area of triangle BCE

$= \frac{1}{2} * \text{Base} * \text{Height}$

$= \frac{1}{2} * BC * EF$

$= \frac{1}{2} * 1 * \frac{\sqrt{2}}{2}$

 © 1999–2016 Manhattan Review

$$= \frac{\sqrt{2}}{4}$$

The correct answer is option B.

Alternate approach:

From symmetry, we can say that EC, when extended would pass through A.

$AC = \sqrt{AB^2 + BC^2} = \sqrt{1+1} = \sqrt{2}$

We know that if the height of two triangles is the same, the ratio of their area equals the ratio of their base.

Thus, we have:

$$\frac{\text{Area of triangle ABC}}{\text{Area of triangle EBC}} = \frac{\text{Base of triangle ABC}}{\text{Base of triangle EBC}}$$

$$= \frac{AC}{CE}$$

$$= \frac{\sqrt{2}}{1}$$

$$\text{Area of triangle ABC} = \frac{\text{Area of square ABCD}}{2}$$

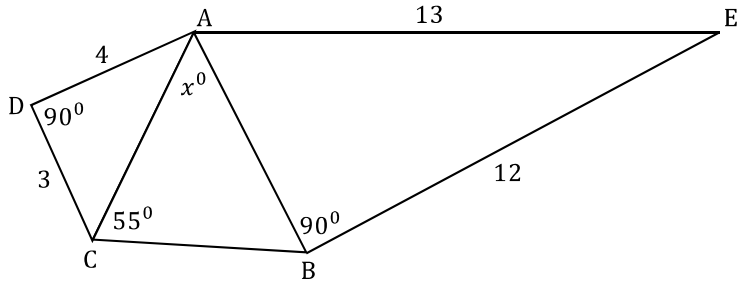
$$\frac{1 * 1}{2} = \frac{1}{2}$$

Thus, we have:

$$\frac{\left(\frac{1}{2}\right)}{\text{Area of triangle EBC}} = \frac{\sqrt{2}}{1}$$

$$\Rightarrow \text{Area of triangle EBC} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$$

215. Let us bring out the figure.



In the right angled triangle ADC, as per Pythagoras theorem, we have:

$AC^2 = AD^2 + CD^2$

$\Rightarrow AC^2 = 3^2 + 4^2 = 25$

$\Rightarrow AC = 5$

In the right angled triangle ABE, as per Pythagoras theorem, we have:

$AB^2 = AE^2 - BE^2$

$\Rightarrow AB^2 = 13^2 - 12^2 = 25$

$\Rightarrow AB = 5$

Thus, we have:

$AB = AC$

In a triangle, if two sides are equal, angles opposite to them are also equal.

$\Rightarrow \angle ABC = \angle ACB = 55^o$

$\Rightarrow \angle BAC = 180^o - (\angle ABC + \angle ACB)$

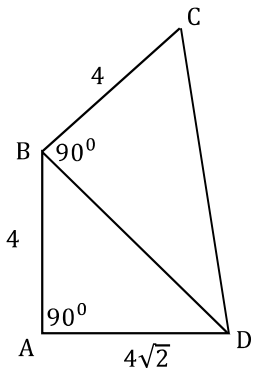
$\Rightarrow \angle BAC = 180^o - (55^o + 55^o)$

$\Rightarrow \angle BAC = 70^o$

$\Rightarrow x = 70$

The correct answer is option D.

216. Let us bring out the figure.



In right angled triangle ABD, as per Pythagoras theorem:

 © 1999–2016 Manhattan Review

$BD^2 = AB^2 + AD^2$

$\Rightarrow BD^2 = 16 + 32 = 48$

$\Rightarrow BD = 4\sqrt{3}$

In right angled triangle CBD, as per Pythagoras theorem:

$CD^2 = CB^2 + BD^2$

$\Rightarrow CD^2 = 16 + 48 = 64$

$\Rightarrow CD = 8$

Thus, perimeter of triangle BCD

$= BC + CD + BD$

$= 4 + 8 + 4\sqrt{3}$

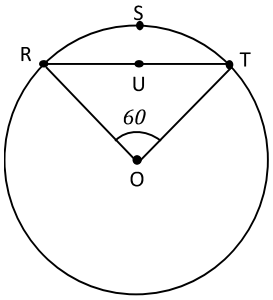
$= 12 + 4\sqrt{3}$

The correct answer is option C.

© 1999–2016 Manhattan Review

5.18 Geometry–Circles

217. Length of the arc RST (which subtends 60° at the center) = $2\pi r * \left(\frac{60}{360}\right)$, where r is the radius of the circle.



Thus, we have:

$2\pi r * \left(\frac{60}{360}\right) = 18\pi$

$=> r = 54$

In triangle TOR, we have: RO = TO = radius of the circle

Hence, we have:

$\angle OTR = \angle ORT = \frac{180^o - 60^o}{2} = 60^o$

Thus, triangle TOR is equilateral.

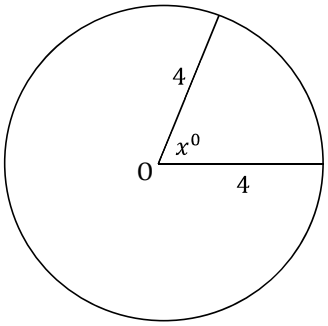
Thus, we have: RT = TO = RO = radius of the circle = 54.

Thus, perimeter of the region RSTU = $(18\pi + 54)$.

The correct answer is option D.

218. Let us bring out the figure.

 © 1999–2016 Manhattan Review



Area of a sector of a circle containing x^0 at the center is given by:

Area $= \pi * (\text{radius})^2 * \left(\frac{x}{360}\right)$

Thus, we have:

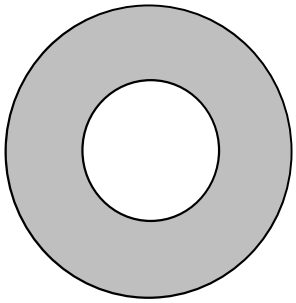
$\pi * (4)^2 * \left(\frac{x}{360}\right) = 2\pi$

$=> x = \frac{2 * 360}{16}$

$= 45$

The correct answer is option C.

219. Let us bring out the figure.



Let the radius of the outer circle be R and the radius of the inner circle be r.

Thus, area of the outer circle = πR^2

Area of the inner circle = πr^2

Thus, area of the shaded region = $(\pi R^2 - \pi r^2)$

Since the area of the shaded region is 3 times the area of the smaller circle, we have:

$\pi R^2 - \pi r^2 = 3\pi r^2$

© 1999–2016 Manhattan Review

$\Rightarrow \pi R^2 = 4\pi r^2$

$\Rightarrow R^2 = 4r^2$

$\Rightarrow R = 2r$

$\Rightarrow \frac{R}{r} = 2$

Thus, we have:

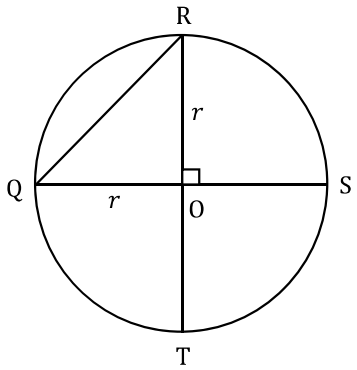
$$\frac{\text{Circumference of the larger circle}}{\text{Circumference of the smaller circle}}$$

$$= \frac{2\pi R}{2\pi r} = \frac{R}{r}$$

$= 2$

The correct answer is option C.

220.



In right angled triangle ROQ, as per Pythagoras theorem, we have:

$\Rightarrow RQ^2 = RO^2 + QO^2$

Since RQ = $\frac{8}{\sqrt{2}}$ and RO = QO = radius = r, we have:

$\Rightarrow \left(\frac{8}{\sqrt{2}}\right)^2 = r^2 + r^2$

$\Rightarrow 2r^2 = 32$

$\Rightarrow r = 4$

Thus, area of the circle

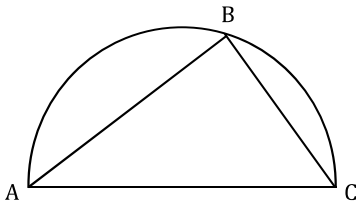
$= \pi r^2$

 © 1999–2016 Manhattan Review

$= 16\pi$

The correct answer is option C.

221. Let us bring out the figure.



AC is the diameter of the semicircle.

Since the diameter subtends 90° at any point on the circumference, in triangle ABC, we have:

$\angle ABC = 90^o$

Thus, from Pythagoras' theorem, we have:

$AC^2 = AB^2 + BC^2$

$\Rightarrow AC^2 = 8^2 + 6^2$ (it is given that AB = 8 and AC = 6)

$\Rightarrow AC = 10$

$\Rightarrow$ Radius of the semicircle $= \frac{10}{2} = 5$

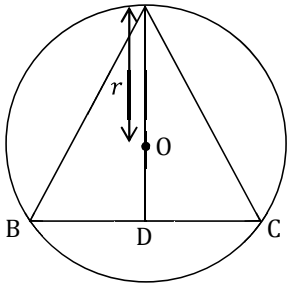
Thus, length of the arc ABC

$= \pi *$ (radius of the circle)

$= 5\pi$

The correct answer is option E.

222.



The area of an equilateral triangle = $\frac{\sqrt{3}}{4} * (\text{side})^2 = \frac{\sqrt{3}}{4} * (\text{AB})^2$

Thus, we have:

$\frac{\sqrt{3}}{4} * (\text{AB})^2 = 9\sqrt{3}$

$\Rightarrow \text{AB} = 6$

Thus, the height (or, the median) of the equilateral triangle (AD)

$= \frac{\sqrt{3}}{2} * (\text{side}) = \frac{\sqrt{3}}{2} * (\text{AB})$

$= \frac{\sqrt{3}}{2} * 6$

$= 3\sqrt{3}$

In an equilateral triangle, the centre of the circumscribing circle is the centroid of the triangle.

Thus, the centre of the circle (O) is the centroid of the equilateral triangle ABC.

The centroid divided the median AD in the ratio 2 : 1.

Thus, we have:

$\text{AO} = \left(\frac{2}{2+1}\right) * \text{AD}$

$= \frac{2}{3} * 3\sqrt{3}$

$= 2\sqrt{3}$

Thus, the radius (r) of the circle = AO = $2\sqrt{3}$

Thus, the area of the circle = πr^2

$= \pi * (2\sqrt{3})^2$

 © 1999–2016 Manhattan Review

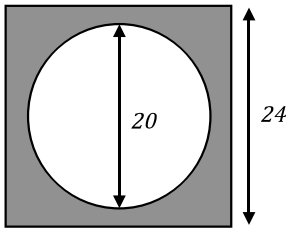
$= 12\pi$

The correct answer is option C.

© 1999–2016 Manhattan Review

5.19 Geometry–Polygon

223. The diagram corresponding to the data provided is shown below:



We need to find what fraction the shaded area represents of the area of square table top.

Side of square table top is 24 inches.

Area of the square tabletop = $(\text{Side})^2$

$\Rightarrow 24^2 = 576$ square inches.

Diameter of circular mat is 20 inches.

Area of the circular mat = $\pi(r)^2$

$\Rightarrow \pi * \left(\frac{20}{2}\right)^2 = \sim 3.14 * 100 = 314$ square inches.

Thus, area of the tabletop not covered by the mat = 576 – 314 = 262 square inches.

Thus, the required fraction = $\frac{\text{area of the tabletop not covered by the mat}}{\text{Area of square table top}}$

$\Rightarrow \frac{262}{576} = \frac{131}{288}$.

We can see that 131 is less than half of 288 (= 144)

So here option A, B, and C are eliminated.

However, 131 is more than one-fourth of 288 (= 72)

So here option D is eliminated.

Thus, the required fraction must be between 50% (half) and 25% (one-fourth).

The only option satisfying is option E which is $\frac{9}{20} * 100 = 45\%$

The correct answer is option E.

 © 1999–2016 Manhattan Review

224. Let the length and breadth of the rectangular floors be x meters and y meters, where x and y are integers.

Since the perimeter of all such floors is 16 meters, we have:

$2(x + y) = 16$

$\Rightarrow x + y = 8$

Thus, the possible cases are:

(1) $(x, y) = (7, 1)$
$\Rightarrow$ Area of the floor $= 1 * 7 = 7$ square meters
$\Rightarrow$ Number of carpets required $= \left(\frac{7}{1 * 1}\right) = 7$
$\Rightarrow$ Total cost of carpeting $= \$(7 * 6) = \$42$

(2) $(x, y) = (6, 2)$
$\Rightarrow$ Area of the floor $= 2 * 6 = 12$ square meters
$\Rightarrow$ Number of carpets required $= \left(\frac{12}{1 * 1}\right) = 12$
$\Rightarrow$ Total cost of carpeting $= \$(12 * 6) = \$72$

(3) $(x, y) = (5, 3)$
$\Rightarrow$ Area of the floor $= 3 * 5 = 15$ square meters
$\Rightarrow$ Number of carpets required $= \left(\frac{15}{1 * 1}\right) = 15$
$\Rightarrow$ Total cost of carpeting $= \$(15 * 6) = \$90$

(4) $(x, y) = (4, 4)$
$\Rightarrow$ Area of the floor $= 4 * 4 = 16$ square meters
$\Rightarrow$ Number of carpets required $= \left(\frac{16}{1 * 1}\right) = 16$
$\Rightarrow$ Total cost of carpeting $= \$(16 * 6) = \$96$

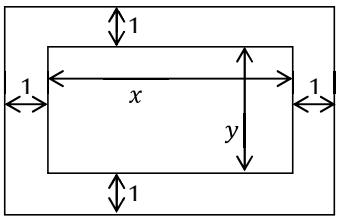
Note: For a given perimeter, the area is the maximum if the length and breadth are equal.

The correct answer is option D.

225. Let the length and breadth of the photograph be x inches and y inches respectively.

Thus, along with the border of 1 inch, the effective length becomes $(x + 2)$ inches and the effective breadth becomes $(y + 2)$ inches (since the border is along all sides) as shown in the figure below:

© 1999–2016 Manhattan Review



Thus, we have:

$(x + 2)(y + 2) = m$

$=> xy + 2(x + y) = m - 4 \ldots$ (i)

Again, along with the border of 2 inches, the effective length becomes $(x + 4)$ inches and the effective breadth becomes $(y + 4)$ inches.

Thus, we have:

$(x + 4)(y + 4) = m + 52$

$=> xy + 4(x + y) = m + 52 - 16$

$=> xy + 4(x + y) = m + 36 \ldots$ (ii)

Subtracting (i) from (ii):

$4(x + y) - 2(x + y) = (m + 36) - (m - 4)$

$=> 2(x + y) = 36 - (-4)$

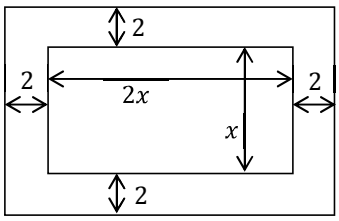
$=> 2(x + y) = 40$

=> Perimeter of the photograph = 40 inches.

The correct answer is option D.

226. Let the breadth of the picture without the border be x inches.

 © 1999–2016 Manhattan Review



Thus, the length of the picture without the border is $2x$ inches.

Thus, area of the above rectangle without the border = $2x * x = 2x^2$ square inches.

Including the border, the length and breadth are $(2x + 4)$ and $(x + 4)$ inches respectively.

Thus, area of the above rectangle including the border = $(2x + 4)(x + 4)$ square inches.

Thus, the area of the border

= (Area of the outer rectangle) − (Area of the inner rectangle)

$= (2x + 4)(x + 4) - 2x^2$

$= 2x^2 + 12x + 16 - 2x^2$

$= 12x + 16$

Thus, we have:

$12x + 16 = 196$

$=> x = 15$

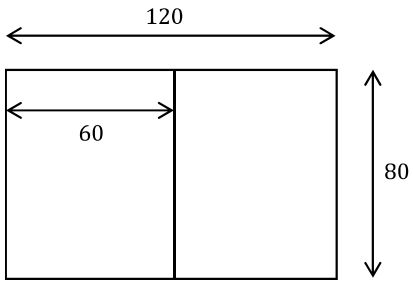
Thus, the length of the picture without the border = $2x = 30$ inches.

The correct answer is option C.

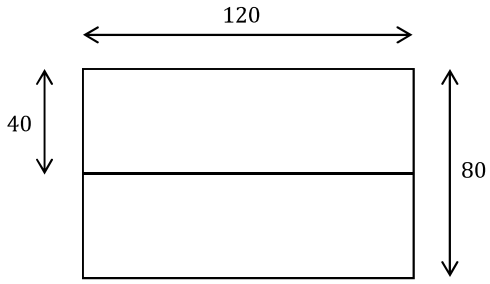
227. The two possible ways in which the field may be partitioned are shown below:

© 1999–2016 Manhattan Review

Case (a)



Case (b)



Case A: Perimeter of one half of the field = $2 * (60 + 80) = 280$ feet.

Case B: Perimeter of one half of the field = $2 * (40 + 120) = 320$ feet.

Thus, the perimeter is the minimum in the first case.

Thus, the minimum cost of fencing = $\$ (280 * 2) = \$560$.

The correct answer is option D.

228. We know that the perimeter is 560 feet.

Thus, twice the sum of the length and breadth is 560 feet.

Thus, the sum of the length and breadth is $\frac{560}{2} = 280$ feet.

Let the length of the field = l feet.

Thus, the breadth of the field = $(280 - l)$ feet.

The diagonal of a rectangle is the square-root of the sum of squares of the length and breadth (from Pythagoras' theorem).

Since the diagonal is 200 feet long, we have:

$\sqrt{l^2 + (280 - l)^2} = 200$

$=> l^2 + (280 - l)^2 = 200^2$

 © 1999–2016 Manhattan Review

$=> 2l^2 + 280^2 - 560l = 200^2$

$=> 2l^2 - 560l + 38400 = 0$

$=> l^2 - 280l + 19200 = 0$

$=> 280l - l^2 = 19200$

$=> l * (280 - l) = 19200$

=> Length * Breadth = 19200

Note: We do not need to solve for the length to get the answer.

Thus, area = 19200 square feet.

The correct answer is option A.

Alternate approach:

We know that the perimeter of the rectangle is 560 feet.

Thus, the sum of the length and breadth is $\frac{560}{2} = 280$ feet.

The area of the rectangle would be the maximum if the length and breadth are equal.

In that case, we would have:

Length = Breadth = $\frac{280}{2} = 140$ feet.

Thus, the maximum possible area would be = 140 * 140 = 19600 square feet.

However, in this case, the diagonal would not be 200 feet (since it does not satisfy Pythagoras' theorem).

Thus, the actual area would be less than 19600 square feet.

There is only one option, 19200 square feet which satisfies the above condition.

229. Width of the strip = t inches

$= \left(\frac{t}{12}\right)$ feet

Length of the strip = m miles

$= 5280 * m$ feet

© 1999–2016 Manhattan Review

Thus, total area of the strip

$$= \left(\frac{t}{12}\right) * (5280 * m)$$

$$= \left(\frac{5280 * m * t}{12}\right) \text{ square feet}$$

Quantity of paint required to paint p square feet = 1 gallon.

Thus, quantity of paint required to paint 1 square foot = $\left(\frac{1}{p}\right)$ gallon.

Thus, quantity of paint required to paint $\left(\frac{5280 * m * t}{12}\right)$ square feet

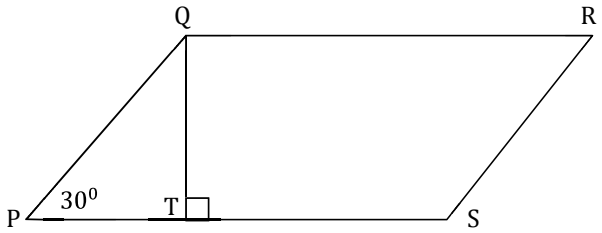
$$= \left(\frac{1}{p}\right) * \left(\frac{5280 * m * t}{12}\right)$$

$$= \left(\frac{5280 * m * t}{12 * p}\right) \text{ gallons}$$

The correct answer is option A.

230. Let us bring out the figure.

 © 1999–2016 Manhattan Review



We know that the area of a parallelogram = Base $*$ Height = PS $*$ QT

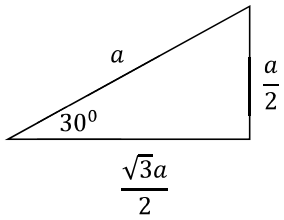
In triangle PQT, we have:

PQ = 4, and

$\angle QPT = 30^o$

In a 30-60-90 triangle, the ratio of the sides of the triangle is shown below:

© 1999–2016 Manhattan Review



If the hypotenuse is of length a:

(1) The side opposite to 30° is $\frac{a}{2}$

(2) The side opposite to 60° is $\frac{\sqrt{3}a}{2}$

In the given problem, we have:

PQ = Hypotenuse = 4

=> QT (side opposite to 30°) = $\frac{4}{2} = 2$

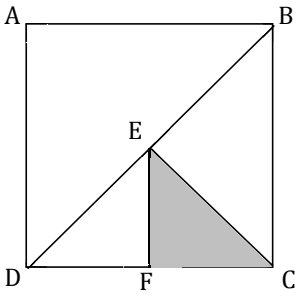
Thus, area of the parallelogram = QT $*$ PS = QT $*$ QR = $2 * 6 = 12$.

The correct answer is option B.

:

231. Let us bring out the figure**.**

 © 1999–2016 Manhattan Review



We know that:

E is the mid-point of BD, and

F is the mid-point of DC.

Thus, from intercept theorem, we have: EF ||BC.

Thus, triangle DEF ≅ triangle DBC

The ratio of the corresponding sides of the above two similar triangles

$= \frac{DE}{DB} = \frac{1}{2}$

Thus, ratio of the area of similar triangles DEF and DBC

$=$ (Raio of their corresponding sides)^2

$= \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

Thus, we have:

$$\frac{\text{Area of triangle DEF}}{\text{Area of triangle DBC}} = \frac{1}{4}$$

Since area of triangle DBC is $\frac{1}{2}$ area of square ABCD, we have:

$$=> \frac{\text{Area of triangle DEF}}{\left(\frac{\text{Area of square ABCD}}{2}\right)} = \frac{1}{4}$$

$$=> 2 * \frac{\text{Area of triangle DEF}}{\text{Area of square ABCD}} = \frac{1}{4}$$

$$=> \frac{\text{Area of triangle DEF}}{\text{Area of square ABCD}} = \frac{1}{8}$$

From the diagram, it is clear that triangles DEF and EFC are congruent and hence, have equal area.

© 1999–2016 Manhattan Review

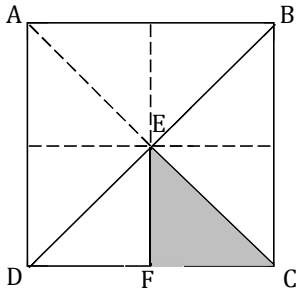
Thus, we have:

$$\frac{\text{Area of triangle EFC}}{\text{Area of square ABCD}} = \frac{1}{8}$$

Alternate approach:

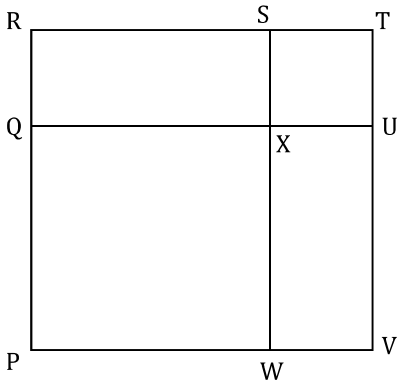
Constructing a few lines (shown below as dotted lines) makes it clear that the shaded part i.e. triangle EFC is 1 out of 8 identical parts in which the square has been divided into.

 © 1999–2016 Manhattan Review



The correct answer is option B.

232. Let us bring out the figure.



Area of square PRTV = 81

Thus, we have:

$RT^2 = 81$

$\Rightarrow RT = 9 \ldots (i)$

Ratio of the area of square region XSTU to the area of square region PQXW = 1 : 4

$\Rightarrow ST^2 : QX^2 = 1 : 4$

$\Rightarrow ST : QX = 1 : 2$

Since RSXQ is a rectangle, RS = QX

$\Rightarrow ST : RS = 1 : 2 \ldots (ii)$

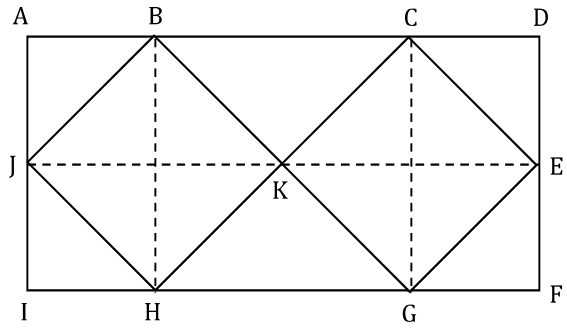
Thus, from (i) and (ii):

© 1999–2016 Manhattan Review

$RS = \left(\frac{2}{1+2}\right) * 9 = 6$.

The correct answer is option C.

233. Let us redraw the figure.



Since BJHK and CKGE are identical squares, we have:

BH = CG = JK = KE (since diagonals of a square are equal)

Thus, we have:

AD = JK + KE

= BH + CG

= AI + AI (since BH = CG = AI)

=> AD = 2 * AI

Since, perimeter of the rectangle is $18\sqrt{2}$, we have:

$2 * (AD + AI) = 18\sqrt{2}$

$=> 2 * (2 * AI + AI) = 18\sqrt{2}$

$=> AI = 3\sqrt{2}$

=> Diagonal of each square = $3\sqrt{2}$

If the side of each square be x, then the length of the diagonal = $x\sqrt{2}$

Thus, we have:

$x\sqrt{2} = 3\sqrt{2}$

 © 1999–2016 Manhattan Review

$=> x = 3$

Thus, perimeter of each square

$= 4x$

$= 4 * 3$

$= 12$

The correct answer is option B.

234. Let the number of desks placed be d and the number of bookshelves placed be b.

Each desk is 2.0 meters long, and each bookshelf is 1.5 meters long.

Thus, the total length covered by d desks and b bookshelves = $(2.0d + 1.5b)$ meters.

We need to find the closest we can take the above value of $(2.0d + 1.5b)$ to 16 (since available space is 16 meters long).

Since we need to maximize the total number of desks and bookshelves, and bookshelves are shorter than desks, we must have a larger number of bookshelves than desks, keeping at least one of each.

Thus, we have:
$(2d + 1.5b) \leq 16$ (Where d and b are positive integers which are not equal to zero)

If we take the maximum value of b to be 10, we have: $1.5b = 1.5 * 10 = 15 < 16$

In the above situation, there is only $(16 - 15) = 1$ meter left which is not sufficient to accommodate a desk.

Checking with the next possible value of $b = 9$, we have: $1.5b = 1.5 * 9 = 13.5 < 16$

Length of space left = $(16 - 13.5) = 2.5$ meters.

Thus, 1 desk (of length 2 meters) can be accommodated in this space.

Thus, space left after accommodating the desk = $(2.5 - 2) = 0.5$ meters.

The correct answer is option A.

235. We know that the radius of the circle formed is r.

Thus, circumference of the circle = $2\pi r$.

© 1999–2016 Manhattan Review

Thus, length of wire left to form the square = $(40 - 2\pi r)$.

Thus, length of each side of the square = $\frac{1}{4}(40 - 2\pi r) = \left(10 - \frac{1}{2}\pi r\right)$.

Thus, total area of the circle and the square formed

$= \pi r^2 + \left(10 - \frac{1}{2}\pi r\right)^2$

The correct answer is option E.

 © 1999–2016 Manhattan Review

5.20 Geometry–3 Dimensional

236. The volume of liquid is the same in both cylinders.

Volume of cylinder = $\pi * r^2 * h$

For first cylinder, we know that $r = 5$ inches and $h = 9$ inches.

Volume of liquid in the first cylinder = $v = \pi * 5^2 * 9 = 225\pi$ cubic inches.

Let the radius of the second cylinder be R inches and height be 4 inches.

Thus, volume of liquid in the second cylinder = $V = \pi * R^2 * 4 = 4R^2\pi$ cubic inches.

As per given data,

$v = V$

Thus, we have:

$4R^2\pi = 225\pi$

$=> R^2 = \frac{225\ \pi}{4\ \pi}$

$=> R = \frac{15}{2} = 7.5$

The correct answer is option D.

237. Diameter of the smaller rim = 28 inches.

Circumference of the smaller rim = Distance covered by smaller ring in one revolution

$=>\ \pi * 28 = 28\pi$ inch.

Number of revolutions made per second by the smaller rim = x.

D_1= distance covered per second by the smaller rim

= No. of revolutions per second $\times$ Distance covered by smaller ring in one revolution

$=> 28\pi x$ inch.

Diameter of the larger rim = 35 inch.

Circumference of the larger rim = Distance covered by larger ring in one revolution

© 1999–2016 Manhattan Review

$\Rightarrow \pi * 35 = 35\pi$ inch.

Let the number of revolutions made per second by the larger rim be y.

D_2 = distance covered per sec. by larger rim

= No. of revolutions per sec. $\times$ Distance covered by larger ring in one revolution

$\Rightarrow 35\pi y$ inch.

Hence, we have:

$D_1 = D_2$

$28\pi x = 35\pi y$

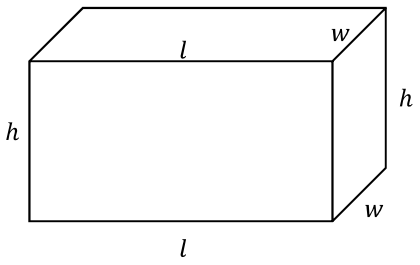
$\Rightarrow y = \frac{4x}{5}$

Thus, the number of revolutions made per second by the larger rim = $\frac{4x}{5}$.

Hence, the number of revolutions made per minute by the larger rim = $\frac{4x}{5} * 60 = 48x$.

The correct answer is option C.

238.



Let the dimensions of the length, width and height be l, w and h respectively.

Since three faces shown have areas 12, 15, and 20, we have:

Top face: $l * w = 12 \ldots$ (i)

Front face: $l * h = 15 \ldots$ (ii)

Right face: $h * w = 20 \ldots$ (iii)

Multiplying the above three equations:

$l^2 * w^2 * h^2 = 12 * 15 * 20 = 3600$

 © 1999–2016 Manhattan Review

$=> l * w * h = \sqrt{3600} = 60$

Since the volume of the solid is given by $l * w * h$, we have:

Volume of the solid = 60.

The correct answer is option A.

239. Since the ratio of length to width to height is 3 : 2 : 2, we can assume that:

- Length = $3k$
- Width = $2k$
- Height = $2k$

Thus, the volume of the rectangular carton

$= 3k * 2k * 2k$

$= 12k^3$

Thus, we have:

$12k^3 = x$

$=> k^3 = \frac{x}{12}$

$=> k = \sqrt[3]{\frac{x}{12}}$

Thus, the height of the carton

$= 2k$

$= 2 * \sqrt[3]{\frac{x}{12}}$

$= \sqrt[3]{8 * \frac{x}{12}}$

$= \sqrt[3]{\frac{2}{3}x}$

The correct answer is option B.

240. Let the height and radius of can X are h and r, respectively.

Thus, volume of oil in can X = $\pi r^2 h$

Thus, price of oil of quantity $\pi r^2 h$ = \$2 ... (i)

Height and radius of can Y = $2h$ and $2r$, respectively.

Thus, volume of can Y = $\pi(2r)^2\,(2h) = 8\pi r^2 h$

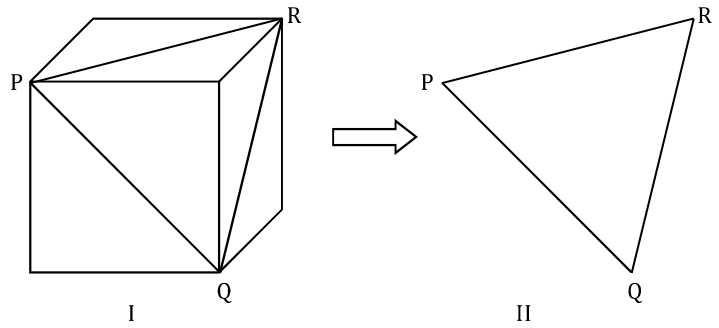
Since can Y is filled to half the capacity, volume of oil in can Y = $\frac{1}{2} * 8\pi r^2 h = 4\pi r^2 h$

Thus, from (i), we have:

Price of oil of quantity $4\pi r^2 h$ = \$(4 * 2) = \$8

The correct answer is option D.

241.



Let us join PR as shown in the figure (I) above.

Let us now remove the edges of the cube from the figure (I) so that only the triangle PQR is left behind as shown in figure (II).

We can see that PR, RQ and PQ are the face diagonals of the cube and hence, are equal to one another.

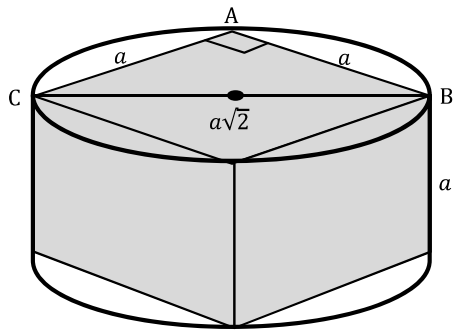
Thus, triangle PQR is an equilateral triangle.

Thus, $\angle PQR = 60^o$

The correct answer is option C.

242. Let us consider the situation where the largest possible cube (shaded in grey) fits perfectly in a particular cylinder as shown in the diagram below; however, the question does not state that we must fit in a largest possible cube into the cylinder.

 © 1999–2016 Manhattan Review



Let the edge of the cube be a.

Since the cube has to have a perfect fit (minimum non-utilization of cylinder space), we must have:

Height of the cylinder = edge of the cube = a

In right angled triangle CAB:

$CB^2 = CA^2 + AB^2 = a^2 + a^2 = 2a^2$

$\Rightarrow CB = a\sqrt{2}$

Thus, the diameter of the cylinder = $a\sqrt{2}$

$\Rightarrow$ Radius of the cylinder = $\frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}$

Thus, volume of the cylinder

$= \pi * \text{radius}^2 * \text{height}$

$= \pi * \left(\frac{a}{\sqrt{2}}\right)^2 * a$

$= \frac{\pi a^3}{2}$

$= \frac{3}{2}a^3$

Volume of the cube = a^3

Thus, volume of the cylinder not occupied by the cube

$= \frac{3}{2}a^3 - a^3$

$= \frac{a^3}{2}$

© 1999–2016 Manhattan Review

Thus, the required percent

$$= \frac{\text{Volume of the cylinder not occupied by the cube}}{\text{Volume of the cylinder}} * 100$$

$$= \frac{\left(\frac{a^3}{2}\right)}{\left(\frac{3}{2}a^3\right)} * 100$$

$$= 33.3\%$$

The above situation depicts the case where minimum volume of the cylinder as a percent of the total volume of the cylinder is unutilized, i.e. not covered by the cube.

Thus, in any other scenario, the required percent value would be either greater than or equal to 33.3%

The only possible value from the answer options is 36%

The correct answer is option E.

 © 1999–2016 Manhattan Review

5.21 Co-ordinate geometry

243. The length of the line segment between 2 points $(x_1,\ y_1) = (-3, -6)$ and $(x_2,\ y_2) = (5,\ 0)$ is given by:

$$L = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(5 - (-3))^2 + (0 - (-6))^2}$$

$$= \sqrt{8^2 + 6^2}$$

$$= 10$$

Thus, the length of the diameter of the circle = 10.

Thus, the length of the radius of the circle = $\frac{10}{2} = 5$.

Thus, the area of the circle = $\pi * 5^2 = 25\pi$.

The correct answer is option C.

244. Equation of a line is: $y = mx + c$, where m is the slope and c is the Y-intercept.

Thus, we have the equation of the given line as:

$y = 2x + 2$

Let the required point on the above line be: $(a, 500)$.

Thus, we have:

$500 = 2a + 2$

$=> a = 249$

Thus, the required X-coordinate of the point is 249.

The correct answer is option A.

245. Equation of a line passing through a point (p, q) and slope m is given as:

$y - q = m\,(x - p)$

Thus, the equation of the line n that passes through the origin $(0,\ 0)$ and has slope 4 is:

© 1999–2016 Manhattan Review

$y - 0 = 4(x - 0)$

$\Rightarrow y = 4x$

Since $(1, c)$ is a point on the line, we have:

$c = 4 * 1$

$\Rightarrow c = 4 \ldots$(i)

Since $(d, 2)$ is a point on the line, we have:

$2 = 4 * d$

$\Rightarrow d = \frac{1}{2} \ldots$(ii)

Thus, from (i) and (ii), we have:

$$\frac{c}{d} = \frac{4}{\left(\frac{1}{2}\right)} = 8$$

The correct answer is option E.

246. The equation of a circle having center at (p, q) and radius r is:
$(x - p)^2 + (y - q)^2 = r^2$

Since the center of the circle is at $(-2, -3)$, the equation of the circle is:
$(x - (-2))^2 + (y - (-3))^2 = r^2$

$\Rightarrow (x + 2)^2 + (y + 3)^2 = r^2$

If a point (m, n) lies inside the circle $(x - p)^2 + (y - q)^2 = r^2$, it must satisfy:

$(m - p)^2 + (n - q)^2 < r^2$

Since $(-2,\ 1)$ lies inside the circle, it must satisfy:
$(-2 + 2)^2 + (1 + 3)^2 < r^2$

$\Rightarrow r^2 > 16$

$\Rightarrow r > 4$ OR $r < -4$

Since r must be positive, we have:
$r > 4 \ldots$(i)

If a point (m, n) lies outside the circle, it must satisfy:

 © 1999–2016 Manhattan Review

$(m-p)^2 + (n-q)^2 > r^2$

Since $(4, -3)$ lies outside the circle, it must satisfy:
$(4+2)^2 + (-3+3)^2 > r^2$

$=> r^2 < 36$

$=> -6 < r < 6$

Since r must be positive, we have:
$0 < r < 6 \ldots$ (ii)

Thus, from (i) and (ii), we have:

$r = 5$

The correct answer is option B.

247. The three vertices are: $(c,\ d)$, $(c,\ -d)$ and $(-c, -d)$

We know that $c < 0$ and $d > 0$

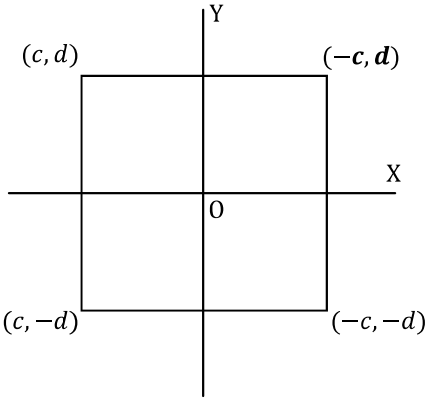
$=> -c > 0$ and $- - d < 0$

We can see that the distance of each of the vertices from the origin $(0,\ 0) = \sqrt{c^2 + d^2}$

Thus, the three vertices are equidistant from the origin.

Thus, the centre of the square is at the origin.

Thus, the square would be positioned as shown in the diagram below:



Thus, the fourth vertex would be $(-c, d)$, which lies in the first quadrant.

© 1999–2016 Manhattan Review

Thus, the X-coordinate of the fourth vertex is positive and the Y-coordinate of the fourth vertex is also positive.

Thus, the only point which also lies in the same quadrant is $(3,\ 5)$.

The correct answer is option E.

248. Distance between two points (x_1, y_1) and (x_2, y_2) is given by: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Distance between the points $(0,\ 0)$ and $(3,\ 3)$

$= \sqrt{(3-0)^2 + (3-0)^2}$

$= \sqrt{18}$

$= 3\sqrt{2}$

Distance between the points $(0,\ 0)$ and $(7,\ 0)$

$= \sqrt{(7-0)^2 + (0-0)^2}$

$= \sqrt{49}$

$= 7$

Distance between the points $(7,\ 0)$ and $(3,\ 3)$

$= \sqrt{(7-3)^2 + (0-3)^2}$

$= \sqrt{25}$

$= 5$

Thus, the perimeter

$= 3\sqrt{2} + 7 + 5$

$= 12 + 3\sqrt{2}$

The correct answer is option E.

249. Since the points are collinear, their slopes must be equal.

Slope of the line joining two points (x_1, y_1) and (x_2, y_2) is given by: $\left(\frac{x_2 - x_1}{y_2 - y_1}\right)$.

Slope of the line joining the points $(a, 0)$ and $(0, b)$

 © 1999–2016 Manhattan Review

$$= \frac{b-0}{0-a}$$

$$= -\frac{b}{a}$$

Slope of the line joining the points $(a, 0)$ and $(1,\ 1)$

$$= \frac{1-0}{1-a}$$

$$= \frac{1}{1-a}$$

Thus, we have:

$$-\frac{b}{a} = \frac{1}{1-a}$$

$$=> -b * (1-a) = a$$

$$=> -b + ab = a$$

$$=> b + a = ab$$

$$=> \frac{b+a}{ab} = 1$$

$$=> \frac{b}{ab} + \frac{a}{ab} = 1$$

$$=> \frac{1}{a} + \frac{1}{b} = 1$$

The correct answer is option C.

250. The area of the triangle formed by the line with the two axes

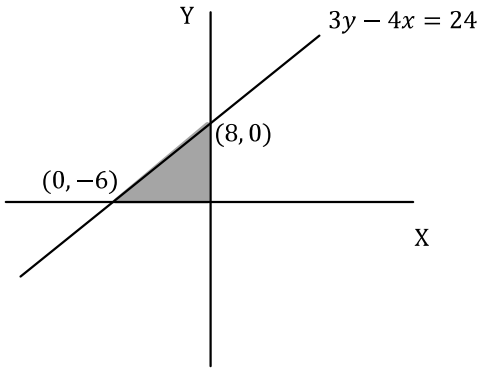
$$= \frac{1}{2} * (\text{Length of X intercept}) * (\text{Length of Y intercept})$$

We have:

$$3y - 4x = 24 \ldots \text{(i)}$$

The graph of the line and the area of the triangle formed by the line and the axes is shown in the diagram below:

© 1999–2016 Manhattan Review



To calculate the X-intercept:

Substituting $y = 0$ in (i):

$x = -\frac{24}{4} = -6$

Thus, the length of the X-intercept = 6.

To calculate the Y-intercept:

Substituting $x = 0$ in (i):

$y = \frac{24}{3} = 8$

Thus, the length of the Y-intercept = 8.

Thus, required area

$= \frac{1}{2} * 6 * 8 = 24$

The correct answer is option C.

 © 1999–2016 Manhattan Review